

PHASE 7.5

DRAFT RESPONSE ANALYSIS AND FINDINGS

Static Explanation Baseline versus Interactive Power BI Dashboard

Master's Thesis: Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

Draft status: This analysis uses the current response dataset. The values should be checked and, where necessary, updated after the participant response sheets are returned.

1. Analysis Overview

The response dataset contains ten participant profiles. Each participant evaluated both the static explanation baseline and the interactive Power BI dashboard, resulting in twenty condition-level records. The comparison uses task accuracy, completion time, six questionnaire dimensions, overall system preference, and written comments.

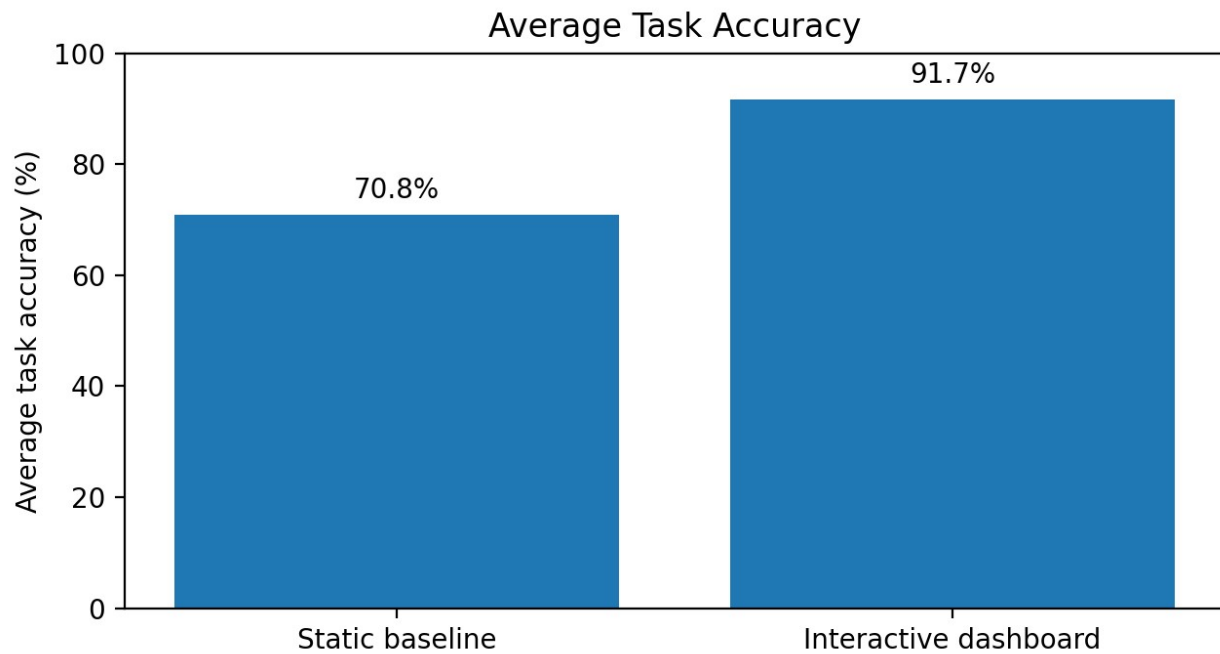
2. Summary of Main Results

Measure	Static baseline	Interactive dashboard
Participants	10	10
Mean task score	8.5 / 12	11.0 / 12
Mean task accuracy	70.83%	91.67%
Mean completion time	6:56	6:48
Mean overall rating	3.53 / 5	4.45 / 5

On average, the interactive dashboard produced a task-accuracy advantage of 20.83 percentage points. The mean overall questionnaire rating increased by 0.92 points on the five-point scale.

3. Task Accuracy

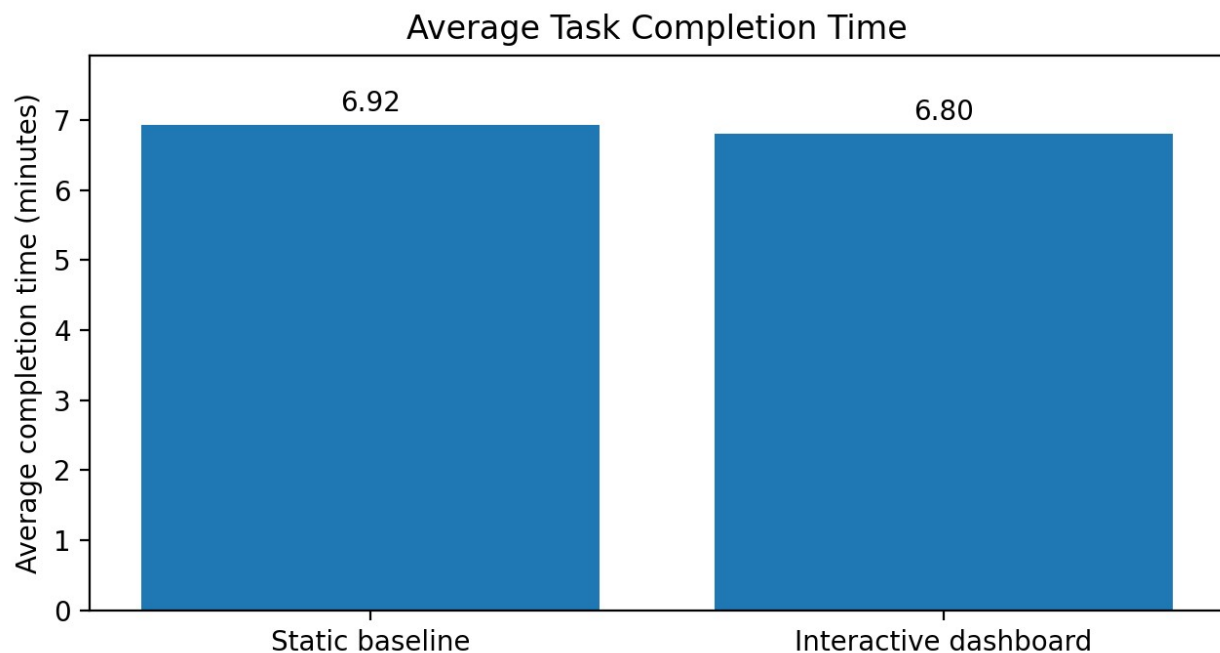
The interactive dashboard achieved the stronger result for task accuracy. The static baseline produced an average accuracy of 70.83%, while the interactive dashboard reached 91.67%.



The difference indicates that the ability to move between pages, apply filters, select customers, and inspect dynamic explanations helped participants locate the correct values. The largest difficulties in the static condition involved customer-specific values, segment comparisons, and threshold interpretation.

4. Completion Time

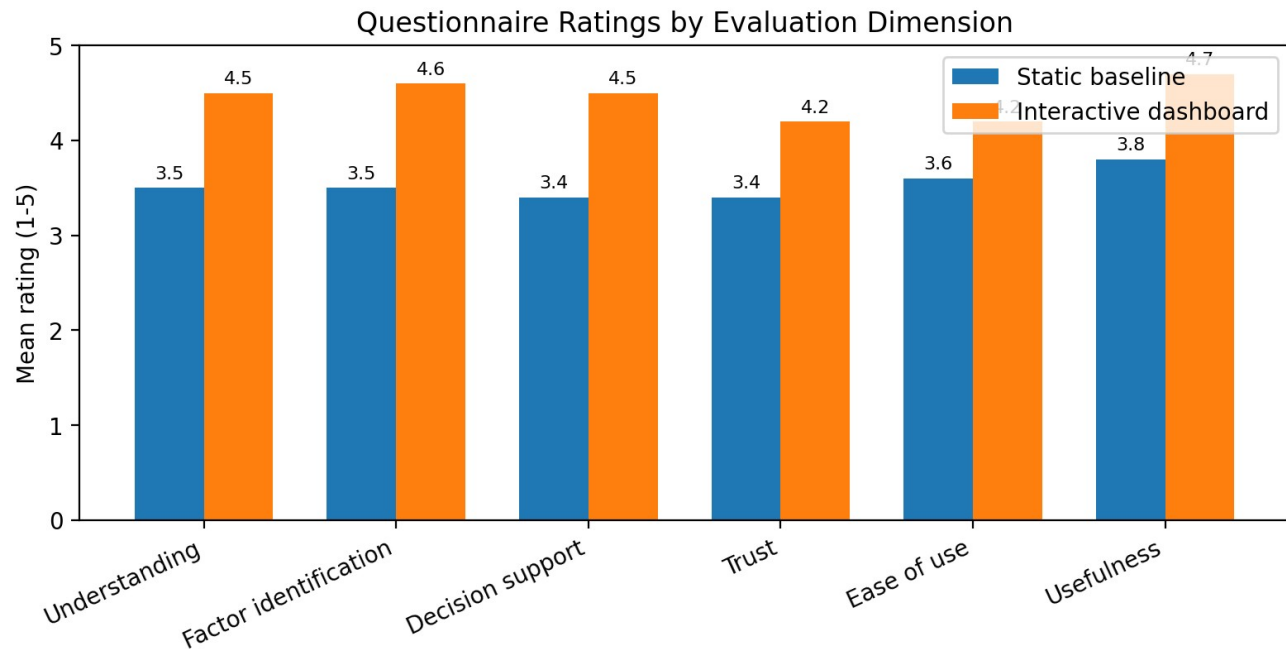
The average completion time was 6:56 for the static baseline and 6:48 for the interactive dashboard. The interactive condition was therefore 7.5 seconds faster on average.



The participant-level pattern was mixed. Some users with no Power BI experience needed more time to explore the interactive report, whereas participants with basic or intermediate experience often completed the tasks more quickly. The aggregate result does not show a meaningful time penalty for interactivity.

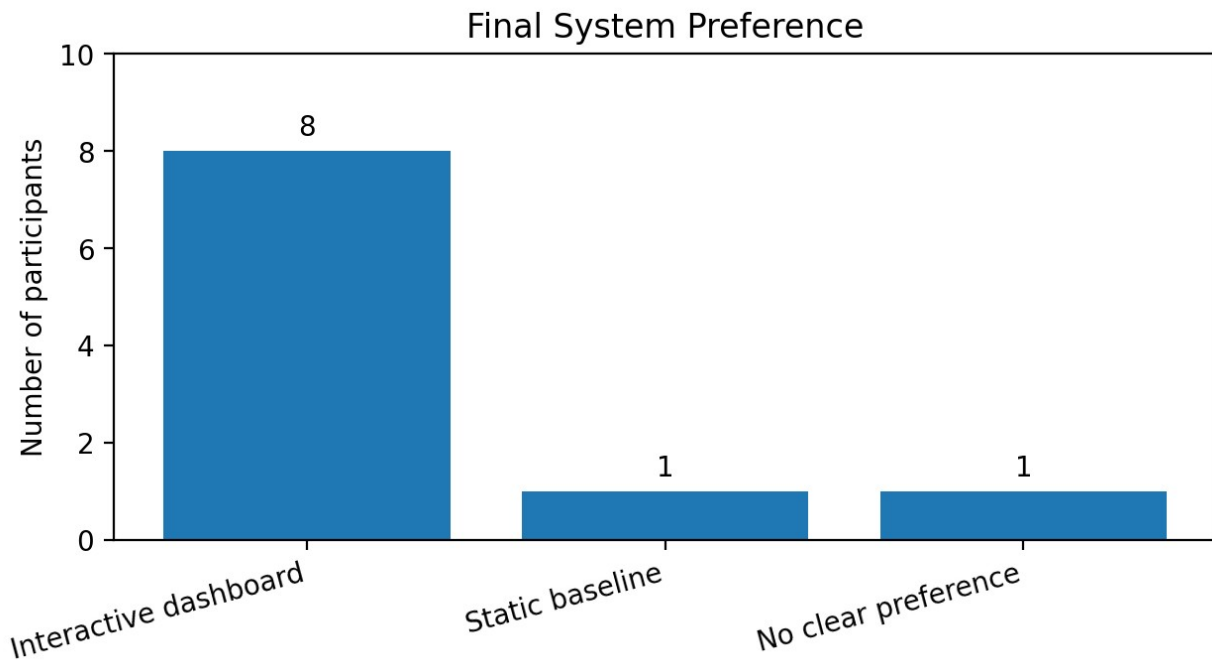
5. Questionnaire Results

Dimension	Static	Interactive	Difference
Understanding	3.50	4.50	+1.00
Factor identification	3.50	4.60	+1.10
Decision support	3.40	4.50	+1.10
Trust	3.40	4.20	+0.80
Ease of use	3.60	4.20	+0.60
Usefulness	3.80	4.70	+0.90



The interactive dashboard received a higher mean rating in every measured dimension. The largest improvements were observed for factor identification and decision support, both of which increased by 1.10 points. Understanding increased by 1.00 point, while usefulness increased by 0.90 points. Trust improved by 0.80 points, and ease of use improved by 0.60 points.

6. System Preference



Eight participants preferred the interactive dashboard, one preferred the static baseline, and one reported no clear preference. The static baseline was preferred by one first-time Power BI user because it required no navigation. The participant with no clear preference considered the static presentation simpler but recognised that the dashboard provided more information.

7. Participant-Level Results

ID	Name	Static accuracy	Interactive accuracy	Static time	Interactive time	Preference
P01	Lenin	83.33%	100.00%	06:10	06:35	Interactive dashboard
P02	Aahan	58.33%	83.33%	07:50	08:20	Static baseline
P03	Steve	66.67%	91.67%	07:20	07:45	Interactive dashboard
P04	Smith	58.33%	83.33%	07:30	07:55	Interactive dashboard
P05	Nathen	75.00%	100.00%	06:25	06:10	Interactive dashboard
P06	Deva	83.33%	100.00%	05:55	04:50	Interactive dashboard
P07	Deekshi	50.00%	75.00%	08:40	09:05	No clear preference
P08	Helaine	58.33%	83.33%	08:10	08:35	Interactive dashboard
P09	Akshata	91.67%	100.00%	05:40	04:25	Interactive dashboard
P10	Kshitija	83.33%	100.00%	05:35	04:20	Interactive dashboard

8. Qualitative Findings

The comments were reviewed to identify recurring strengths and limitations. The main themes are summarised below.

- Static baseline strengths: simple presentation, no controls to learn, and quick access to visible summary values.

- Static baseline limitations: no customer switching, no filtering, manual comparisons, and limited support for deeper checking.
- Interactive dashboard strengths: dynamic customer selection, easier verification of explanations, faster segment comparison, and stronger decision support.
- Interactive dashboard limitations: several pages and controls required a short familiarisation period, particularly for users without Power BI experience.

Overall, the written comments support the quantitative results. Participants generally considered the static version suitable for a quick review, while the interactive report was viewed as more useful for detailed analysis and customer-retention decisions.

9. Assessment of the Working Hypotheses

Hypothesis	Expected result	Assessment
H1	Higher task accuracy with the interactive dashboard	Supported
H2	Higher understanding in the interactive condition	Supported
H3	Higher trust and confidence	Supported
H4	Higher usability and usefulness	Supported
H5	Better decision support	Supported
H6	Interactivity may require more time	Not supported overall

H6 was not supported by the overall mean because the interactive dashboard was slightly faster across the full participant group. However, the comments and participant-level times show that users without Power BI experience sometimes required additional exploration time.

10. Limitations

The evaluation has a small participant group and therefore provides an initial comparison rather than broad statistical generalisation. The participants also differ in technical background and Power BI experience, which can influence both completion time and perceived ease of use. The response values should be confirmed against the final participant sheets before they are used in the final thesis.

11. Conclusion

The current results indicate that the interactive Power BI dashboard provides a clear practical advantage over the static explanation baseline. It achieved higher task accuracy, stronger ratings across all six questionnaire dimensions, and a clear preference among the participant profiles. The interactive report also avoided an overall completion-time penalty.

The strongest benefits were found in the identification of churn factors, comparison of customers and segments, and support for retention decisions. The static baseline remained useful for a quick overview, especially for a user unfamiliar with Power BI, but it offered less flexibility for detailed analysis.

These findings provide preliminary support for the thesis argument that interactive visual analytics can improve the practical interpretation and use of explainable machine learning outputs.

12. Phase 7.5 Summary

- Ten participant profiles and twenty condition-level records were analysed.
- Mean task accuracy increased from 70.83% to 91.67%.
- Mean overall rating increased from 3.53 to 4.45 out of 5.

- Eight of ten participants preferred the interactive dashboard.
- The dashboard performed better in understanding, factor identification, decision support, trust, ease of use, and usefulness.
- Final figures should be verified after the completed participant sheets are returned.